Model comparison

testing parameters, and choosing between models

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Data Analysis for Cognitive Neuroscience





Two ways to test a hypothesis

Within one model

a test on a **parameter**

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Same question, asked two different ways.

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- **more complex models** — no formula at all, or one nobody wants to derive

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- the price is computation: you fit the model hundreds of times

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- **Non-nested models** — neither is a special case of the other: different sets of regressors, or different model families altogether

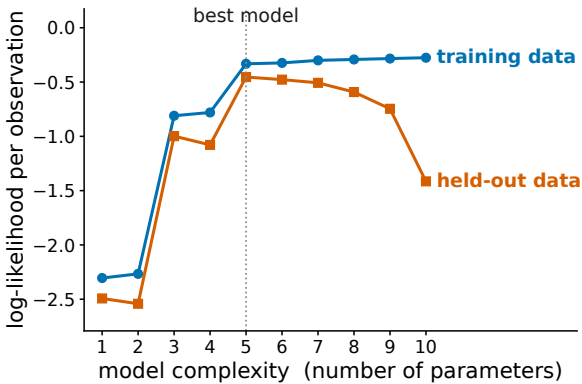
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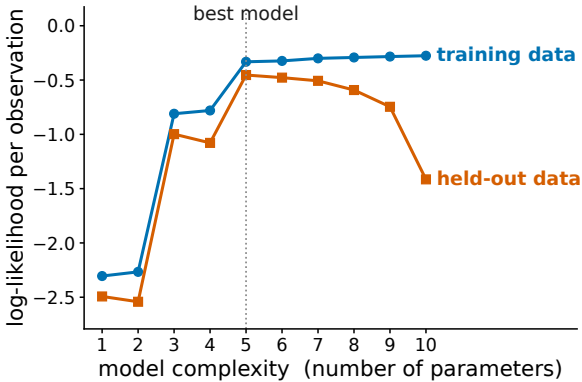


- testing w_2 and comparing these two models are the *same question*
- **Non-nested models** — neither is a special case of the other: different sets of regressors, or different model families altogether
- only the model-comparison framework can handle those

Why not simply compare the maximum likelihood?

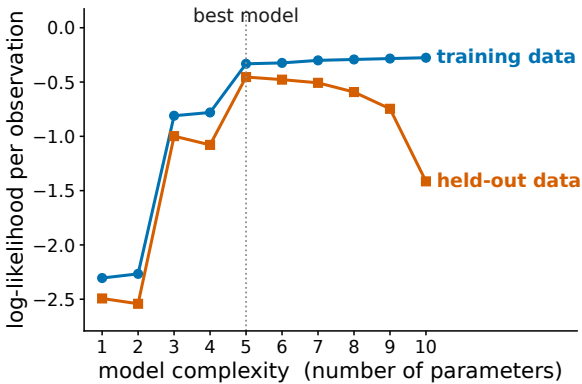


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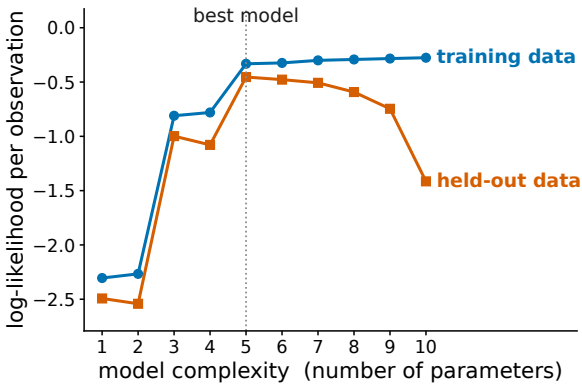
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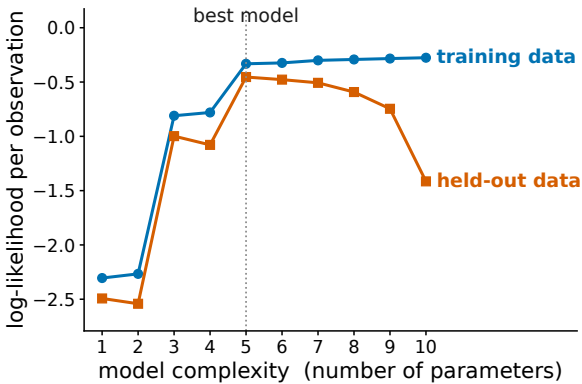
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So the raw likelihood cannot arbitrate.

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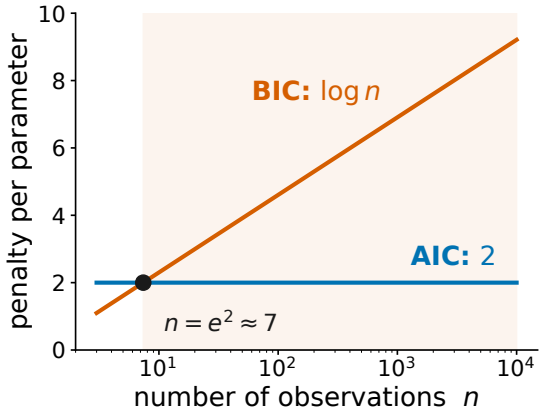
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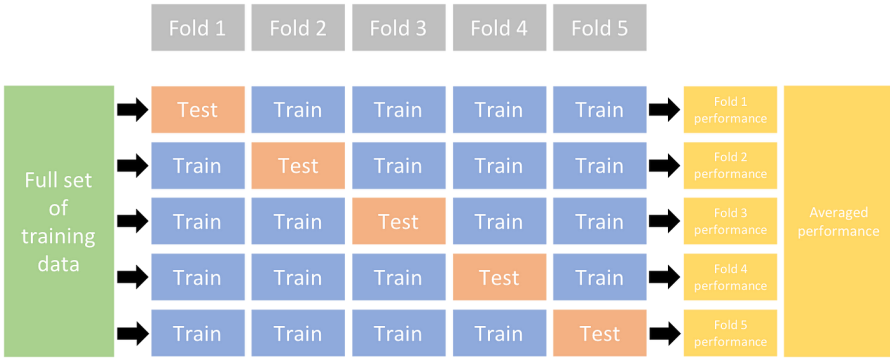
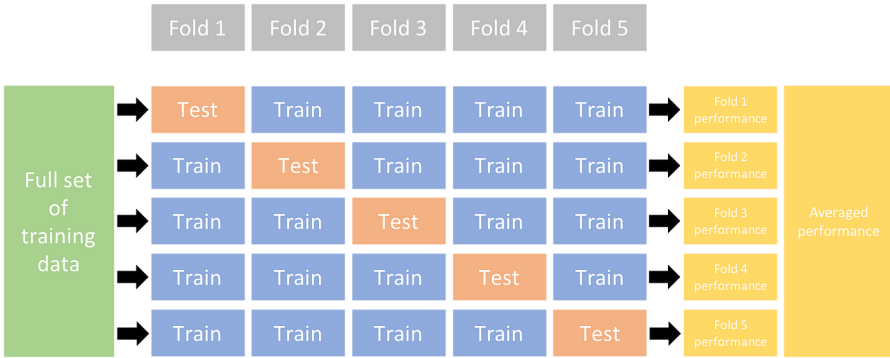


Figure from docs.ultralytics.com/guides/kfold-cross-validation

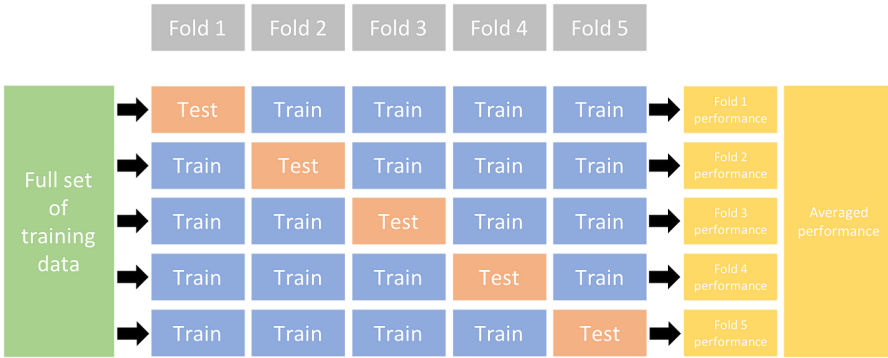
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- every fold takes its turn as the test set, and the scores are summed: **cross-validation**

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Which one to use?

	AIC / BIC	Cross-validation
to compute	+ a one-line formula	– a loop over folds
computing time	+ one fit	– K fits per model
justification	– an approximation, valid asymptotically	+ measures directly what we care about